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AUTOMOTIVE TRIM RETAINING SYSTEM WITH DIVIDED HOUSING

Abstract

A retaining system for automotive trim includes a component that has a hole, another component that has a housing, and a retainer that engages with the housing and with the hole. The housing includes a receiver having a cap defining an opening and the housing defines an entry. The opening is configured to receive the retainer when the retainer is inserted through the entry. A clip or clips is/are disposed at the entry and is/are configured to flex to allow insertion of the retainer through the entry. The housing is divided by a gap that is defined between the clip(s) and the cap.

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Background/Summary

INTRODUCTION

[0001] The present disclosure generally relates to connecting one component to another and more particularly relates to systems with mating retainers and housings, where the housing has a balance of strength and ergonomically beneficial compliance.

[0002] In manufactured articles, many approaches may be taken to connect one component to another. Generally, some type of fastening, bonding, fusion, and/or coupling is used. The approach selected is determined by the available access, surface quality considerations, strength requirements, ergonomic considerations, and other criteria. For example, components may be screwed, bolted, welded, glued, nailed, stapled, riveted, or otherwise fitted and secured together.

[0003] Automotive door trim panels serve as an interface with the vehicle's interior and exist between the interior and the sheet metal structure of, and inner mechanisms of, the door. The interior facing side may be referred to as an A-surface and has an aesthetic requirements, while the structure facing side may be referred to as the B-surface, which is generally not visible in the completed product. The door trim panels typically consist of a substrate covered with trim material. These door panels are connected with the door's structure by releasable retainers. The retainers are separate components that connect with a housing of the trim panel and then connect with the door structure, often with some type of snap-in fit that is completed in a blind fashion behind the B-surface. The system is preferably strong enough to resist all the forces that it may experience during the operation of the vehicle door and also ergonomic by allowing an ease insertion and extraction of the retainer.

[0004] Accordingly, it is desirable to provide mating components with a retainer system that is strong and ergonomic for use in various applications. Furthermore, other desirable features and characteristics of the present invention will become apparent from the subsequent detailed description and the appended claims, taken in conjunction with the accompanying drawings and the foregoing introduction.

SUMMARY

[0005] A retaining system such as for automotive trim includes, in a number of embodiments, a component that has a hole, another component that has a housing, and a retainer that engages with the housing and with the hole. The housing includes a receiver having a cap defining an opening and the housing defines an entry. The opening is configured to receive the retainer when the retainer is inserted through the entry. A clip or clips is/are disposed at the entry and is/are configured to flex to allow insertion of the retainer through the entry. The housing is divided by a gap that is defined between the clip(s) and the cap.

[0006] In additional embodiments, the housing includes a receiver disposed at a distance from a surface of the second component and connected to the surface of the second component by a wall. The receiver operates to receive and hold the retainer.

[0007] In additional embodiments, the clip is one of a pair of clips, where the pair of clips define the entry. The opening has a diameter and the entry has a width. The diameter is larger than the width when the pair of clips are in a relaxed state.

[0008] In additional embodiments, the clip is one of a pair of clips. The pair of clips define the entry. The opening has a diameter, and the entry has a width, and the diameter is larger than the width when the pair of clips are in a relaxed state.

[0009] In additional embodiments, the retainer includes a shaft, a head and a flange. The housing includes a platform, with supports extending from the platform toward the cap. The receiver defines a slot between the cap and the platform. The slot operates to receive the head and the opening operates to receive the shaft.

[0010] In additional embodiments, the housing includes a platform, with supports extending from the platform toward the cap, and a plurality of ribs are formed on the cap for strength.

[0011] In additional embodiments, the housing includes a platform, and the clip includes a tab that

operates to flex the clip when a force is applied to the tab toward the platform.

[0012] In additional embodiments, the housing includes a platform. The clip includes a tab that operates to flex the clip when a force is applied to the tab toward the platform. A stop extends from the platform toward the tab. The stop operates to limit flexure of the clip to an amount within an elastic range.

[0013] In additional embodiments, the housing is produced within a form that has a pair of die elements that part at the gap. The section is disposed at the gap and operates to leave a membrane on the housing at the gap.

[0014] In additional embodiments, the clip operates to flex between a relaxed state and a flexed state. The housing operates to receive the retainer when the clip is in the flexed state and operates to hold the retainer in the housing when the clip is in the relaxed state.

[0015] In a number of additional embodiments, one component has a hole, and a housing is included on another component. A retainer operates to engage with the housing of the other component and with the one component at the hole. The housing includes a receiver that has a cap defining an opening and the housing defines an entry. The opening operates to receive the retainer when the retainer is inserted through the entry. A pair of clips is disposed at the entry. The clips define the entry and operate to flex to allow insertion of the retainer through the entry. The housing is divided by a gap that is defined between the clips and the cap.

[0016] In additional embodiments, the housing includes a receiver disposed at a distance from a surface of the second component and the receiver is connected to the surface of the second component by a wall. The receiver operates to receive and to releasably hold the retainer.

[0017] In additional embodiments, the clips define the entry. The opening has a diameter that is substantially the same in size as a shaft of the retainer. The entry has a width, and the diameter is larger than the width, when the clips are in a relaxed state.

[0018] In additional embodiments, the opening has a diameter, and the entry has a width between the clips. The diameter is larger than the width, when the first clip and the second clip are in a relaxed state.

[0019] In additional embodiments, the retainer includes a shaft, a head and a flange. The housing includes a platform, with supports extending from the platform toward the cap. The receiver defines a slot between the cap and the platform. The slot operates to receive the head, and the opening operates to receive the shaft. The supports operate to guide the head through the slot and to assist in flexing of the first and second clips.

[0020] In additional embodiments, the housing includes a platform, with supports extending from the platform toward the cap. The supports operate to support the retainer, and a plurality of ribs are formed on the cap for strength.

[0021] In additional embodiments, the housing includes a platform, and the clips include cantilevered legs. The clips each include a tab extending opposite the respective cantilevered leg. The tabs operate to effect flexing of the clips when forces are applied to the tabs toward the platform.

[0022] In additional embodiments, the housing includes a platform. The clips each include a tab configured to flex the respective clip when a force is applied to the respective tab toward the platform. A stop extends from the platform toward each of the first and second tabs. The stops operate to limit flexure of the clips to an amount within an elastic range.

[0023] In additional embodiments, the housing is produced within a form that has a pair of die elements that part at the gap. The section is disposed at the gap and operates to leave a membrane on the housing at the gap.

[0024] In a number of other embodiments, a retention system for a trim component of a vehicle includes a base component that has a hole. The base component is a panel of the vehicle. A housing is defined on the trim component. A retainer operates to engage with the housing of the trim component and with the base component at the hole. The housing includes a receiver that has a cap

defining an opening, and the housing defines an entry. The opening operates to receive the retainer when the retainer is inserted through the entry. The housing includes a pair of clips that are disposed at, and define, the entry. The housing is divided by a first gap that is defined between one clip and the cap and by a second gap that is defined between the other clip and the cap. The gaps enable the clips to flex relative to the cap to allow insertion of the retainer through the entry and into the cap.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The exemplary embodiments will hereinafter be described in conjunction with the following drawing figures, wherein like numerals denote like elements, and wherein:

[0026] FIG. 1 is a schematic diagram of a vehicle with a component retainer system, in accordance with various embodiments;

[0027] FIG. 2 is a perspective illustration of a housing for the retainer system of FIG. 1, in accordance with various embodiments;

[0028] FIG. 3 is a schematic, sectioned illustration of part of the retainer system of FIG. 1 with a retainer in the housing, in accordance with various embodiments;

[0029] FIG. 4 is a schematic diagram of a retainer of the retainer system of FIG. 1 with a pin and grommet, in accordance with various embodiments;

[0030] FIG. 5 is a schematic illustration of the housing of the retainer system of FIG. 1 illustrating compliance action, in accordance with various embodiments;

[0031] FIG. 6 is a schematic illustration of the housing of the retainer system of FIG. 1 illustrating another compliance action, in accordance with various embodiments;

[0032] FIG. 7 is a perspective illustration of a housing for the retainer system of FIG. 1, in accordance with various embodiments;

[0033] FIG. 8 is a plan view of a housing for the retainer system of FIG. 1, in accordance with various embodiments;

[0034] FIG. 9 is a perspective illustration of a housing for the retainer system of FIG. 1, in accordance with various embodiments;

[0035] FIG. 10 is a fragmentary, perspective illustration of a housing for the retainer system of FIG. 1 showing an element of a mold, in accordance with various embodiments; and

[0036] FIG. 11 is a fragmentary, sectional illustration of a housing and molding die for the retainer system of FIG. 1, in accordance with various embodiments.

DETAILED DESCRIPTION

[0037] The following detailed description is merely exemplary in nature and is not intended to limit the application and uses. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding introduction, brief summary or the following detailed description.

[0038] With reference to FIG. 1, illustrated is one example of a retaining system 20 in the context of a vehicle 22 having a body 24. As depicted in FIG. 1, the body 24 of the vehicle 22 may be supported on wheel assemblies (not shown), such as by a suspension assembly (not shown). The body 24 may be of various types that define a physical shape for desired purposes. The body 24 substantially encloses components of the vehicle 22, defines exterior surfaces, and defines an interior 26. The wheel assemblies may each be rotationally coupled near a respective corner of the body 24. In various embodiments, the vehicle 22 may be a front wheel drive vehicle, a rear wheel drive vehicle, or an all-wheel drive vehicle with any number of wheels, however other drive arrangements are contemplated.

[0039] Propulsion of the vehicle 22, such as on a roadway, may be provided by a propulsion system

(not shown). The propulsion system may be any of various types such as electric, internal combustion, hybrid, or others. Torque generated by the propulsion system is transferred to the wheel assemblies to propel the vehicle **22**, which may be done under manual and/or autonomous control.

[0040] In the embodiment illustrated in FIG. **1**, the interior **26** of the vehicle **22** includes a base component **30**, which in this embodiment is an interior structural panel of a door of the vehicle **22**. A trim component **32** is configured to cover the base component **30** and to face the interior **26** of the vehicle **22**. The base component **30** may be a door panel as shown, a trunk panel, a roof panel, or any other component, including in non-vehicular applications, to which the trim component **32** may be attached. The trim component **32** may be, without limitation, an interior trim panel for a door as shown, a trunk trim panel, a roof trim panel, or any other component, including in non-vehicular applications, for connection with the base component **30**.

[0041] In the assembly of articles, one of the challenges in efficiently connecting together of components may involve connecting a part, such as the trim component **32** to another component, such as the base component **30**, where the fasteners are invisible from the outer surface **36** of the assembled trim component **32**. According to a number of variations described in relation to FIG. **1**, the trim component **32** may be connected to the base component **30** by a number of retainers, represented by the retainer **40**. The retainer **40** of the retaining system **20** is connected with the trim component **32** as described below and is configured to engage the trim component **32** through one or more holes **42**. Features and components shown in other figures may be incorporated and used with those shown in FIG. **1**.

[0042] Referring to FIGS. **2** and **3** along with FIG. **1**, the back surface **38** of the trim component **32**, which is a side opposite the outer surface **36**, includes a number of housings, represented by the housing **44**. The housing **44** is configured to receive and hold the retainer **40**, such as shown in FIG. **3**. The housing **44** is constructed to provide strength for securely holding the retainer **40**, while also having compliance to be receptive to releasably receiving the retainer **40**. The housing **44** may be referred to as a “doghouse” due to its shape and appearance. The housing **44** provides a number of functions including as a convenient structure to releasably connect with the retainer **40** and to locate the retainer **40** at a projected location from the back surface **38** of the trim component **32**.

[0043] The housing **44** may include a base **46** that is connected to the trim component **32** at the back surface **38** and extends away from the back surface **38** to a receiver **48**. The receiver **48** is presented a distance away from the back surface **38** and includes an opening **50** for receiving and holding the retainer **40**. The receiver **48** is connected with, and is supported on the trim component **32** by, a wall **52**. The wall **52** forms the base **46** and as such is connected with the trim component **32**. The wall **52** has an open side **54**. In the current embodiment, the wall **52** is curved around the area between the receiver **48** and the back surface **38** from an edge **56** to another edge **58**. In other embodiments, the wall **52** may take another shape such as semi-cubic, angled, and others. The open side **54** is bounded by the edges **56**, **58**, the receiver **48** and the back surface **38**. The space **60** inside the wall **52** is substantially open.

[0044] In general, the receiver **48** includes a platform **62**, a cap **64**, at least one clip (clips **66** and **68**), supports **70** and ribs **72**. The receiver **48** defines the opening **50**, gaps **74**, **76** and an entry **78**. The platform **62** is wall-like and is joined with the wall **52** at its end **80** opposite the base **46** and defines a side of the space **60**. The supports **70** extend from the platform **62** and act as slides/ramps for guiding the retainer **40** into the opening **50**. The supports **70** also work to provide support during assembly of the door trim component **32** to the sheet metal of the base component **30**. When the trim component **32** is pushed to the base component **30**, the pin and grommet (retainer **40**) needs a support underneath to help it to be push inside the hole **42**. In the current embodiment, there are three supports **70** although any number may be used. The cap **64** may be flat or may have the retainer's shape depending on the design intent, and in embodiments may be shaped to conform

to the shape of the retainer **40** and to capture the retainer **40** by the receiver **48**. As shown in FIG. 3, the opening **50** receives a shaft **82** of the retainer **40** and the cap **64** contains a head **84** of the retainer **40**. As such, the retainer **40** is held by the housing **44**. A flange **86** is provided on the shaft **82** with a wall **88** of the cap **64** of the housing **44** of the trim component **32** captured between the head **84** and the flange **86**. The ribs **72** are provided around the outside of the receiver **48** for added strength and may be sized, shaped and located as needed. The ribbing is optional and in some cases the housing **44** doesn't need the ribbing, therefore, in embodiments the ribs **72** may not be provided for the housing **44**.

[0045] The clips **66** and **68** are disposed on the same side of the housing **44** as the open side **54**. The clips **66** and **68** define the entry **78** which is aligned with, and leads to, the opening **50**. In some embodiments, only one clip **66**, or **68** may be used. For example, only the clip **66** may be present with the cap **64** extending on the other side of the opening **78** where the clip **68** is absent. The gap **74** is defined between the clip **66** and the cap **64**. The clip **66** has a leg **90** that extends from the platform **62** and a leg **92** that extends toward the clip **68**. Similarly, the gap **76** is defined between the clip **68** and the cap **64**. The clip **68** has a leg **94** that extends from the platform **62** and a leg **96** that extends toward the clip **66**. The entry **78** is defined between the legs **94** and **96**. The clips **66** and **68** have an angled shape and are cantilevered from the platform **62** so that they may flex under force with the legs **92** and **96** moving at various angles to vary the size/width of the entry **78** for insertion and/or removal of the retainer **40** from the receiver **48**. In embodiments, the width of each clip **66**, **68** closer to the platform **62** may be larger than the width near the legs **92**, **96**. The shape could be angled but also straight or with any shape as long it allows the clips **66**, **68** to bend. The gaps **74** and **76** create a “divided” structure of the receiver **48** where the cap **64** provides retaining strength and the clips **66** and **68** provide a tunable way of receiving and maintaining the retainer **40** in the opening **50** for ergonomics and effectiveness. The amount of force needed to insert the retainer **40** may be adjusted by the design of the clips **66** and **68** including by the selection of material used, their thickness and width, and their structural features such as ribs and contours.

[0046] Referring to FIG. 4, details of an embodiment of the retainer **40** are shown. In this embodiment, the retainer **40** is a pin-and-grommet type. In other embodiments, the retainer may be another type such as threaded, push-in (e.g. tree or bird-beak shaped), or others, with or without a grommet. As illustrated, the retainer **40** includes a pin **100** and a grommet **102**. The pin **100** is configured for retention in the housing **44** of the trim component **32** and the grommet **102** is configured for retention in the hole **42** of the base component **30**. The grommet **102** includes a plug **104** for insertion into the hole **42** and a flange **106** at one end of the plug **104**. A through-hole **108** is formed through the flange **106** and the plug **104** for receiving the pin **100**. The pin **100** includes the shaft **82**, the head **84** and the flange **86**. The shaft **82** extends from the opposite side of the flange **86** from the head **84** to a tip **114** for insertion into the grommet **102**. Between the flange **86** and the tip **114**, the shaft **82** includes a ridge **110** and a narrowed section **112** for clipping into the grommet **102**. During assembly, the pin **100** is inserted into the housing **44** with the tip **114** projecting away from the back surface **38** of the trim component **32** and the grommet **102** is inserted into the hole **42** of the base component **30**. With the pin **100** held by the housing **44**, the tip **114** is inserted into the through hole **108** and the trim component **32** is retained on the base component **30**. In other embodiments, the housing **44** may be on the base component **30** and the retainer **40** may be held thereon, and the grommet **102** or hole **42** may be on the trim component **32**.

[0047] Referring to FIG. 5, the action of the clips **66**, **68** is schematically illustrated. The clips **66**, **68** are shown in their relaxed state in solid-line form and in their flexed state in dashed-line form. In the relaxed state, the distance between the supports **70** and the clips **66**, **68** creates and interference to hold the retainer **40** in the opening **50**. In their flexed state, the size/width of the entry **78** is increased as compared to the relaxed state so that the shaft **82** of the retainer **40** fits through. The clips **66**, **68** move with the legs **92**, **96** moving line-to-line and the legs **90**, **94** bending at angles as depicted. Once the shaft **82** is positioned in the opening **50**, the clips **66**, **68** return to the

relaxed state and the retainer **40** is captured in the housing **44**. The profile shape of the clips **66**, **68** may take a variety of forms such as curved, angled, multiple angled, or a combination thereof. In an alternative embodiment of FIG. **6**, the legs **92**, **96** may bend as shown in dashed line form to increase the size of the entry **78**.

[0048] Referring to FIG. **7**, and embodiment of the housing **44** includes release tabs **120**, **122** on the clips **66**, **68**. The release tab **120** projects from the clip **66** at a junction **124** between the leg **90** and the leg **92**. The release tab **120** projects from the junction **124** in an opposite direction from the leg **92**. As a result, a force **126** on the release tab **120** flexes the clip **66** so that the leg **92** moves laterally as the leg **90** bends. Similarly, a force **128** on the release tab **122** flexes the clip **68** so that the leg **96** moves laterally as the leg **94** bends. As a result, the tabs **120**, **122**, via the forces **126**, **128**, may be used to flex the clips **66**, **68** increasing the size/width of the entry **78**. This action may be used to remove the retainer **40** from the housing **44** and may also be used to insert the retainer **40** into the housing **44** in an ergonomically friendly manner. The housing **44** also includes stops **130**, **132** which are positioned between the tabs **120**, **122** and the platform **62**. The stops **130**, **132** extend from the platform **62** toward the release tabs **120**, **122**, and act to limit travel of the tabs **120**, **122** under the forces **126**, **128** to ensure the clips **66**, **68** remain in their elastic range and to avoid plastic deformation or fracture of the clips **66**, **68**.

[0049] Referring to FIG. **8**, details of the receiver **48** of the housing **44** are shown. The outline **140** of the shaft **82** is shown as a reference size that is approximately the same as the diameter **142** of the opening **50** but smaller while still closely fitting within the opening **50** without substantial clearance. In embodiments, the opening **50** may be larger with a clearance given between the diameters per design intent. Notable is that the outline **140** of the shaft **82**, when seated in the opening **50**, is inboard from the clips **66**, **68**. The entry **78** has a width **144** that is smaller than the outline **140** of the shaft **82** so that the clips **66**, **68** hold the retainer **40** in the opening **50** when in their relaxed states. The clips **66**, **68** have a feature at the outboard side (entrance) of the entry **78** to facilitate insertion of the retainer **40**. The feature is an angled configuration that tapers from its widest at the outboard side and narrows moving inward toward the opening **50**. In embodiments, the angle **145** is preferably between 15 and 30 degrees. The clips **66**, **68** have a feature at the inboard side (exit) of the entry **78** tailored to provide a desired amount of retention of the retainer **40** in the housing **44** and to provide an ergonomically friendly extraction of the retainer **40** from the housing **44**. In embodiments to maximize retention, the angle **146** may be ninety degrees. In embodiments where ease of extraction is desired, the angle **146** may be an acute angle. In embodiments, the angle **146** may be selected so that a projecting line **148** extends from the corner **150** of the clip **68** to the corner **152** of the cap **64** at the opening **50**. This geometry avoids binding when insertion and/or extraction of the retainer **40** is undertaken.

[0050] Referring to FIG. **9**, a perspective illustration of the housing **44** shows features of the receiver **48**. The legs **92**, **94** are positioned at the same height from the platform **62** and the supports **70** as the cover of the cap **64**. In some embodiments, the legs **92**, **94** may be positioned closer to the platform **62** and to the supports **70** than the cover **156** of the cap **64**. The result, in such an embodiment, is a step at area **158** across the gaps **74**, **76** and an increased action of the clips **66**, **68** to retain the retainer **40** in the housing **44**. The supports **70** have smoothly rounded corners **162** to guide the head **84** of the retainer **40** into the receiver **48** through the slot **159**. The shape of the corners **162** may be rounded as shown, or angled to provide a ramp-like structure that helps flex the clips **66**, **68** as the retainer **40** is inserted.

[0051] As shown in FIGS. **10** and **11**, when the housing **44** is formed, such as by injection molding, or another process, a mold, die, form or other forming device **168** may be used. The forming device **168** may have die elements that may include a core **164** and a lifter **166**, which is a moveable part. There may be a parting line between the core **164** and the lifter **166** at the gaps **74**, **76**. A section **160** may be used at the part of the forming device **168** that forms the gaps **74**, **76** (gap **74** illustrated). The section **160** may be a part of the core **164** that leaves a membrane **165** at the

parting line. For example, the section **160** may be configured to leave a thin wall, for example a 0.5 mm wall thickness, which may be referred to as the membrane **165**. The section **160** may be disposed in the forming device **168** and may avoid metal-to-metal contact of sections of the forming device **168** that create the gaps **74**, **76** at the parting line. The section **160** facilitates forming the geometry of the housing **44**. After forming, the membrane **165** may flex or tear in service, or be removed from the housing **44**.

[0052] Accordingly, a retaining system with a divided housing is provided to enable secure retention of two components together with a retainer that is selectively added or removed from one of the components that carries the housing. The retaining system may be used in blind assembly where the retainers are hidden behind one of the components as it is assembled to the other component.

[0053] While at least one exemplary embodiment has been presented in the foregoing detailed description, it should be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or exemplary embodiments are only examples, and are not intended to limit the scope, applicability, or configuration of the disclosure in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing the exemplary embodiment or exemplary embodiments. It should be understood that various changes can be made in the function and arrangement of elements without departing from the scope of the disclosure as set forth in the appended claims and the legal equivalents thereof.

Claims

1. A retaining system comprising: a first component having a hole; a second component having a housing; and a retainer configured to engage with the housing of the second component and with the first component at the hole, wherein the housing comprises a receiver having a cap defining an opening and the housing defining an entry, wherein the opening is configured to receive the retainer when the retainer is inserted through the entry, wherein a clip is disposed at the entry, the clip being configured to flex to allow insertion of the retainer through the entry, wherein the housing is divided by a gap that is defined between the clip and the cap.
2. The retaining system of claim 1, wherein: the housing includes a receiver disposed at a distance from a surface of the second component and connected to the surface of the second component by a wall, and the receiver is configured to receive and hold the retainer.
3. The retaining system of claim 1, wherein: the clip is one of a pair of clips, wherein the pair of clips define the entry, the opening has a diameter and the entry has a width, and the diameter is larger than the width when the pair of clips are in a relaxed state.
4. The retaining system of claim 1, wherein: the clip comprises one of a pair of clips, the pair of clips define the entry, the opening has a diameter, and the entry has a width, and the diameter is larger than the width when the pair of clips are in a relaxed state.
5. The retaining system of claim 1, wherein: the retainer includes a shaft, a head and a flange, the housing includes a platform, with supports extending from the platform toward the cap, the receiver defines a slot between the cap and the platform, and the slot is configured to receive the head and the opening is configured to receive the shaft.
6. The retaining system of claim 1, wherein: the housing includes a platform, with supports extending from the platform toward the cap, and a plurality of ribs are formed on the cap for strength.
7. The retaining system of claim 1, wherein: the housing includes a platform, and the clip includes a tab configured to flex the clip when a force is applied to the tab toward the platform.
8. The retaining system of claim 1, wherein: the housing includes a platform, the clip includes a tab configured to flex the clip when a force is applied to the tab toward the platform, and a stop

extending from the platform toward the tab, the stop configured to limit flexure of the clip to an amount within an elastic range.

9. The retention system of claim 1, wherein the housing is configured to be produced within a form that has a pair of die elements that part at the gap, and comprising a section disposed at the gap that is configured to leave a membrane on the housing.

10. The retention system of claim 1, wherein the clip is configured to flex between a relaxed state and a flexed state, wherein the housing is configured to receive the retainer when the clip is in the flexed state and is configured to hold the retainer in the housing when the clip is in the relaxed state.

11. A retaining system comprising: a first component having a hole; a second component having a housing; and a retainer configured to engage with the housing of the second component and with the first component at the hole, wherein the housing comprises a receiver having a cap defining an opening and the housing defining an entry, wherein the opening is configured to receive the retainer when the retainer is inserted through the entry, wherein a first clip and a second clip are disposed at the entry, the first clip and the second clip defining the entry and configured to flex to allow insertion of the retainer through the entry, wherein the housing is divided by a gap that is defined between the first and second clips and the cap.

12. The retaining system of claim 11, wherein: the housing includes a receiver disposed at a distance from a surface of the second component and the receiver connected to the surface of the second component by a wall, and the receiver is configured to receive and to releasably hold the retainer.

13. The retaining system of claim 11, wherein: the first clip and the second clip define the entry, the opening has a diameter that is substantially the same in size as a shaft of the retainer, wherein the entry has a width, and the diameter is larger than the width when the first and second clips are in a relaxed state.

14. The retaining system of claim 11, wherein: the opening has a diameter, and the entry has a width between the first and second clips, and the diameter is larger than the width when the first clip and the second clip are in a relaxed state.

15. The retaining system of claim 11, wherein: the retainer includes a shaft, a head and a flange, the housing includes a platform, with supports extending from the platform toward the cap, the receiver defines a slot between the cap and the platform, and the slot is configured to receive the head, the opening is configured to receive the shaft, and the supports are configured to guide the head through the slot and to assist in flexing of the first and second clips.

16. The retaining system of claim 11, wherein: the housing includes a platform, with supports extending from the platform toward the cap, the supports are configured to support the retainer, and a plurality of ribs are formed on the cap for strength.

17. The retaining system of claim 11, wherein: the housing includes a platform, the first and second clips include cantilevered legs, and the first and second clips each include a tab extending opposite the respective cantilevered leg, the respective tab configured to effect flexing of the respective first and second clip when forces are applied to the tabs toward the platform.

18. The retaining system of claim 11, wherein: the housing includes a platform, the first and second clips each includes a tab configured to flex the respective first and second clip when a force is applied to the respective tab toward the platform, and a stop extending from the platform toward each of the first and second tabs, the respective stop configured to limit flexure of the respective clip to an amount within an elastic range.

19. The retention system of claim 11, wherein the housing is configured to be produced within a form that has a pair of die elements that part at the gap, and comprising a section disposed at the gap that is configured to leave a membrane as a part of the housing.

20. A retention system for a trim component of a vehicle, the retention system comprising: a base component having a hole, wherein the base component comprises a panel of the vehicle; a housing

defined in the trim component; and a retainer configured to engage with the housing of the trim component and with the base component at the hole, wherein the housing comprises a receiver having a cap defining an opening, and the housing defining an entry, wherein the opening is configured to receive the retainer when the retainer is inserted through the entry, wherein the housing includes a first clip and a second clip that are disposed at, and define, the entry, wherein the housing is divided by a first gap that is defined between the first clip and the cap and by a second gap that is defined between the second clip and the cap, and wherein the first gap and the second gap enable the first clip and the second clip to flex relative to the cap to allow insertion of the retainer through the entry and into the cap.
